

30-Day Learning Roadmap (Beginner-Friendly)

Week 1: Foundations

Milestone: Understand basics and key concepts.

- 1 Day 1: Introduction and use cases
- 2 Day 2: Basic terminology
- 3 Day 3: Setup tools (Python, VS Code, Colab)
- 4 Day 4: Learn core concepts
- 5 Day 5: Practice exercises
- 6 Day 6: Watch tutorials and take notes
- 7 Day 7: Revision and quiz

Resources: YouTube, freeCodeCamp, official docs

Mini Project: Simple calculator or text analyzer

Week 2: Core Skills

Milestone: Apply concepts practically.

- 1 Day 8: Intermediate concepts
- 2 Day 9: Practice problems
- 3 Day 10: Real examples
- 4 Day 11: Learn libraries/tools
- 5 Day 12: Build scripts
- 6 Day 13: Debug and improve
- 7 Day 14: Revision

Resources: GitHub, LeetCode, HackerRank

Mini Project: Automation script or preprocessing tool

Week 3: Real-World Application

Milestone: Build a real-world project.

- 1 Day 15: Choose problem
- 2 Day 16: Plan project
- 3 Day 17: Build basic version
- 4 Day 18: Add features
- 5 Day 19: Improve output/UI
- 6 Day 20: Testing
- 7 Day 21: Finalize

Resources: Stack Overflow, Docs, Tutorials

Major Project: Resume screening system or chatbot

Week 4: Mastery & Portfolio

Milestone: Showcase and refine skills.

- 1 Day 22: Optimize project
- 2 Day 23: Add features
- 3 Day 24: Documentation
- 4 Day 25: Upload to GitHub
- 5 Day 26: Learn deployment
- 6 Day 27: Share on LinkedIn
- 7 Day 28: Feedback
- 8 Day 29: Revision
- 9 Day 30: Reflection

Resources: GitHub guides, blogs

Final Project: Polished project ready for portfolio

Final Outcome

- 1 Strong foundation
- 2 2–3 projects completed
- 3 GitHub portfolio
- 4 Confidence to continue learning